

GOURAV KUMAR OJHA

AI/ML Developer

+91 6207001498 | kumargouravojha@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI/ML Developer with hands-on experience architecting production-grade GenAI applications, cloud-native microservices, and distributed data pipelines. Proficient in Python (Pandas, NumPy, Scikit-learn), Go (working knowledge), TypeScript, LLM-driven workflows (LangChain, RAG), FastAPI, Docker, Kubernetes, and CI/CD on GCP. ISRO research collaborator, 3x Hackathon Winner, and co-founder of a defence-tech AI startup. Delivered 60% efficiency gains and 98% accuracy across enterprise-grade AI systems.

EDUCATION

B.Tech in Computer Science (Artificial Intelligence).

Aug 2023 – Present

Rungta College of Engineering and Technology, Bhilai

GPA: 7.8/10 | Relevant Coursework: Data Structures & Algorithms, OOP, Software Engineering, Cloud Computing, Machine Learning, Distributed Systems

WORK EXPERIENCE

Research Collaborator – Martian Dust Storm Analysis |

Aug 2025 – May 2026

Indian Institute of Remote Sensing (ISRO), Dehradun | Remote

- Achieved 90–95% dust storm detection accuracy, by architecting a distributed, cloud-native Python data pipeline (Pandas, NumPy) processing 10TB+ across MRO/MCS, Curiosity (REMS/SAM), and Hubble datasets, validated with unit and integration tests using boundary analysis and equivalence class partitioning across 5 instrument sources.
- Reduced multi-instrument data reconciliation errors, by debugging ML model outputs in Jupyter notebooks and implementing structured failure-recovery and root-cause-analysis workflows across 4 Martian years of atmospheric data.

Co-Founder & AI Engineer – AstraTech (Defence-Tech Startup, Team of 3) |

Jan 2026 – Jul 2026

Rungta Business Incubation RIU, Bhilai | Remote

- Improved sensor-fault detection speed by 30–40%, by architecting a microservices-based, security-first embedded AI system with real-time sensor fusion (LWIR, mmWave, seismic) and a distributed CI/CD release-management pipeline with automated test stages and observability logging.
- Reduced deployment defects across releases, by applying full-SDLC agile delivery and design-thinking-driven requirements analysis across hardware/software teams for role-based access and end-to-end workflow validation; design patent filed (Patent Pending).

KEY PROJECTS

SetuDrishhti – AI-Powered Clinical Decision Support System [\[Live\]](#)

- Achieved 98% diagnostic accuracy, by architecting a full-stack, cloud-native NLP and multi-modal deep learning platform (PyTorch, HuggingFace, LangChain, FastAPI, PostgreSQL, Docker) with RAG-based EHR retrieval and role-based authentication built as independently deployable microservices.
- Reached 85% model accuracy across patient cohorts, by building a distributed SHAP/LIME evaluation harness measuring output quality, defect detection rate, and consistency.

LUMEN AI – Enterprise Financial Assistant [\[GitHub\]](#)

- Cut test turnaround by 60%, by architecting a distributed, multi-agent RAG microservices pipeline on GCP managed services (Vertex AI, BigQuery) with version-controlled prompt deployment and full observability tracing.
- Reduced root-cause-analysis time by 10%, by building a security-first, cloud-native structured logging and tracing layer for transparent multi-agent execution.

HireMind AI – Enterprise Candidate Intelligence Platform [\[Live\]](#)

- Delivered ~34ms startup and ~120ms retrieval latency at 12.5 RPS / 320MB peak memory, by architecting a full-stack, containerized microservices platform (Python, FastAPI, FAISS, Docker) using Hybrid Semantic Retrieval and a deterministic Explainability Engine.
- Cut deployment vulnerabilities by 30%, by implementing a security-first, multi-stage Docker release-management pipeline with Prometheus observability and distributed monitoring.

GuardGen – Hybrid Extractive-Generative Research Assistant [\[Github\]](#)

- Produced citation-grounded academic research drafts rated 100% plagiarism-free and 86–90% AI-content-free, by architecting a hybrid extractive-generative pipeline that prioritizes real human-written academic sources over free-form LLM generation.
- Bridged the gap between raw LLM output and responsible academic writing, by combining retrieval-based extraction with generative synthesis to structurally ground every claim in verified sources.

Multi-Tenant WooCommerce-on-Kubernetes Provisioning Platform [\[Github\]](#)

- Enabled on-demand provisioning of fully isolated, quota-enforced WooCommerce stores on Kubernetes, by capping each tenant at 5 pods, 500m/2-core CPU request/limit, 512Mi/2Gi memory request/limit, 2 PVCs, and 10Gi storage, with complete automated cleanup on deletion.
- Prevented resource-exhaustion incidents cluster-wide, by implementing per-container LimitRanges (100m/200m CPU and 128Mi/256Mi memory default request/limit, 1 core/1Gi max) so containers without explicit limits still receive safe default caps.

TECHNICAL SKILLS

Languages: Python, Go (working knowledge), TypeScript, JavaScript, C/C++, SQL

ML / AI: LangChain, Prompt Engineering, LLM Workflows, Agentic AI, RAG, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Hugging Face, OpenCV, SHAP, LIME, Jupyter Notebooks

Cloud & DevOps: GCP (Vertex AI, BigQuery, Cloud Functions — Managed Services), Docker, Kubernetes (multi-tenant orchestration, quota enforcement, LimitRanges), Microservices Architecture, Distributed Computing, Cloud-Native Applications, CI/CD, Release Management, Git, Prometheus

Web & DB: REST API Development, FastAPI, Flask, Full Stack Development (React, Next.js), PostgreSQL, MongoDB, Vector DBs

Engineering Practices: Security-First Mindset, Design Thinking, Test Automation, Boundary Analysis, End-to-End Workflow Validation, Root Cause Analysis

CERTIFICATIONS

Coding Spoon – Generative AI, Agentic Systems & MLOps Deployment (Jun 2026) | Postman API Expert | Google Cloud Skills Boost – GCP & Applied ML (2025)

Microsoft Azure Fundamentals (AZ-900) | AICTE AI & ML Fundamentals

- Research Intern – Supernova Cosmology Project, India Space Week (Jun–Jul 2025): Estimated Hubble Constant from JWST/HST Type Ia Supernovae data; published findings on Hubble tension.
- Open Source Contributor, GSSoC 2024 (Oct–Nov 2024): Built ML modules across multiple repos, delivering 15–20% performance improvements.

ACHIEVEMENTS

- 1st Place – E-Summit Startup Pitching Competition, Vyom Cultural Fest, RISU (Mar 2026): Won top prize pitching AstraTech in a Shark Tank-style competition, awarded by Padma Shri Anand Kumar.
- 1st Place – Hack-Sprint Hackathon, GDG on Campus RCET & Infinity Coders (Mar 2026): Led Team Alien X to build an AI-powered real-time Sepsis & Deterioration Monitoring System in a 12-hour sprint.
- 3x Regional AI Hackathon Winner — SetuDrishiti, HireMind AI, Blockchain Voting System (2024–present).

PUBLICATIONS (UNDER REVIEW)

“Interannual Evolution of Martian Dust Activity at Gale Crater: A Multi-Instrument Analysis Using Curiosity Rover and MCS Observations (MY32–MY36),” Planetary and Space Science, Elsevier (Under Review, 2026), in collaboration with Indian Institute of Remote Sensing (ISRO).